Previous Differential Equations

We previously found solutions to two classes of differential equations:

- $\frac{dy}{dt} = f(t)$, which we solve by _____ $f(t)$.
- $\frac{dy}{dt} = ky$, which has solution $y(t) =$ _____.

Today, we will solve equations of the form

$$\frac{dy}{dt} = \frac{f(t)}{g(y)}.$$

Note that all the above equations are of this form!

The Idea...

Questions

1. Find $\frac{dy}{dt}$ if $\ln y - \frac{1}{2}t^2 = 17$.

2. By reversing the process we used above, let's solve $\frac{dy}{dt} = -\frac{t}{y}$.

Separation of Variables

Let's generalize the process above. Suppose

$$\frac{dy}{dt} = \frac{f(t)}{g(y)}.$$

(In the second problem above, what is $g(y)$?) Also, let $G(y)$ be an antiderivative of $g(y)$. Then,

$$\frac{d}{dy}G(y) = _____$$

and

$$\frac{d}{dt}G(y(t)) = _____ = _____$$

Notice, that we can rewrite our differential equation as $g(y)\frac{dy}{dt} = f(t)$. Can we rewrite the left hand side as something else?

Therefore, $G(y) + C = \int f(t)\,dt$ (Why?). Or, to put it another way,

$$\boxed{\int g(y)\,dy = \int f(t)\,dt}$$

In the differential equation $\frac{dy}{dt} = \frac{f(t)}{g(y)}$, our formula says that we can "cross-multiply," separating out the dy from the dt in the process, then drawing an integral sign. This is a great strength of the notation, but also a great weakness. On the one hand, it's easy to remember, but on the other hand, it's easy to confuse this mnemonic device with actual mathematical content. Don't do so!

Questions

1. Solve the following initial value problems:
   - $\frac{dy}{dt} = 2 - y$, $y(0) = 3$

- $\frac{dy}{dt} = 2 - y$, $y(0) = 1$

- $\frac{dy}{dt} = y \sin t$, $y(0) = 4$

- $\frac{dy}{dt} = \frac{y^2}{t}$, $y(1) = 2$

2. (a) Solve $\frac{dy}{dx} = ky$ in two ways (k is a constant).

   (b) Check that $y = 0$ is a solution to this differential equation. To what value of the constant does this correspond?

   (c) But if $y = e^B e^{kt}$, then how can $e^B = 0$? Something went wrong here. Why did we lose a solution when we used separation of variables?